

21CSE453T Speech Recognition

Activity-Unit-3

From Voice to Text and Back (Complete Understanding of ASR and TTS)

Objective: Use audio file having stammer (or stutter)/ hiccups.

(Should involve a temporary disruption in normal vocal pattern)

Follow the pattern and do the analysis:

**Speech → Feature Extraction → Log-Mel Spectrum → ASR → CTC → Text
→ TTS → Speech**

Report Pattern:

- Python/Colab notebook
- Waveform and log-Mel spectrogram
- ASR architecture diagram
- CTC explanation
- WER calculations
- ASR comparison table
- TTS output
- 2-page mini-report
- 3-minute demonstration